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- The diagram illustrates the cellular mechanism of Antidiuretic Hormone (ADH) in a collecting duct cell. ADH molecules (represented by black dots) bind to a specific receptor protein on the apical (lumen-facing) membrane. This binding triggers an intracellular signal transduction pathway, labeled 'A', which involves an 'enzyme cascade' that activates 'adenylyl cyclase'. Adenylyl cyclase then converts 'ATP' into 'cAMP' (labeled 'B'). This second messenger system leads to the insertion and opening of water channels (aquaporins) in the apical membrane, allowing 'water' to move from the 'lumen of collecting duct' into the cell. Simultaneously, the enzyme cascade activates another pathway (labeled 'C') that opens water channels in the basolateral membrane, allowing 'water' to exit the cell into the 'blood' (labeled 'P'). The overall effect is the reabsorption of water from the duct lumen into the bloodstream.

(a) Name structures **A**, **B** and **C**.

[3]

- Suggest reasons why the water potential of the blood may decrease.

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- (c) Diabetes insipidus is a condition affecting osmoregulation by the kidney. One form of diabetes insipidus is caused by a tumour in the pituitary gland, which results in a decreased secretion of ADH.

Suggest the symptoms that would occur in a person with diabetes insipidus.

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- (d) Neurogenic diabetes insipidus (NDI) is another form of diabetes insipidus. In NDI, ADH molecules cannot bind to the receptor proteins located in the cell surface membranes of the cells of the collecting duct.

With reference to Fig. 1.1, explain the effect on the cell surface membrane labelled **P** if ADH cannot bind to the receptor proteins.

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- (e) NDI is caused by a recessive allele of the gene coding for the receptor protein. The gene is located on the X chromosome.

Explain why a man with NDI could **not** have inherited the condition from his father.

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[Total: 12]